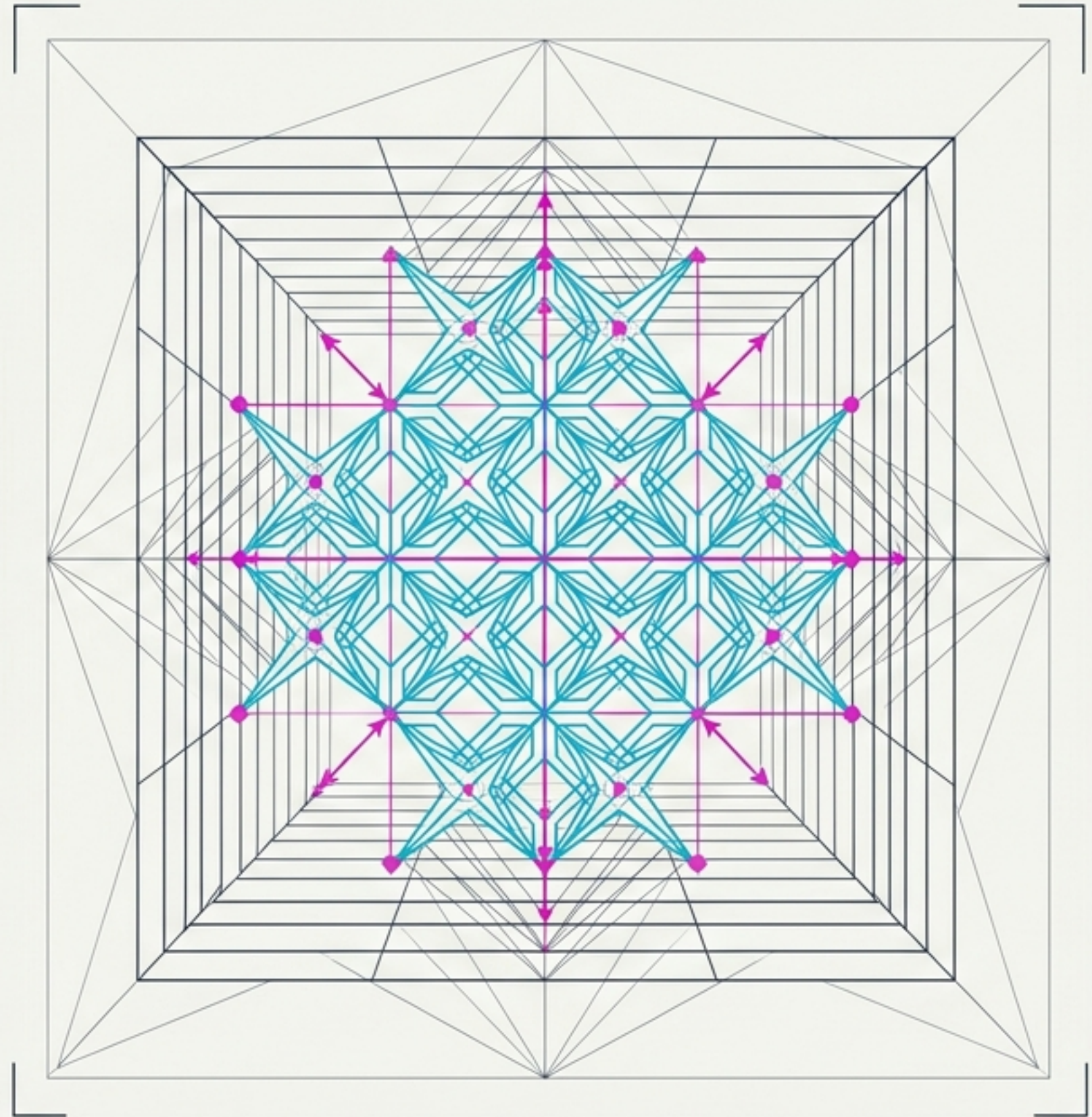


Shattering a 93-Year-Old Geometric Mystery

On September 23, 2026, an OpenAI preprint struck down one of discrete geometry's longest-standing questions: Borsuk's Conjecture. By combining symmetric matrices, topological spheres, and discrete combinatorics, researchers finally found the exact threshold where our geometric intuition breaks down.

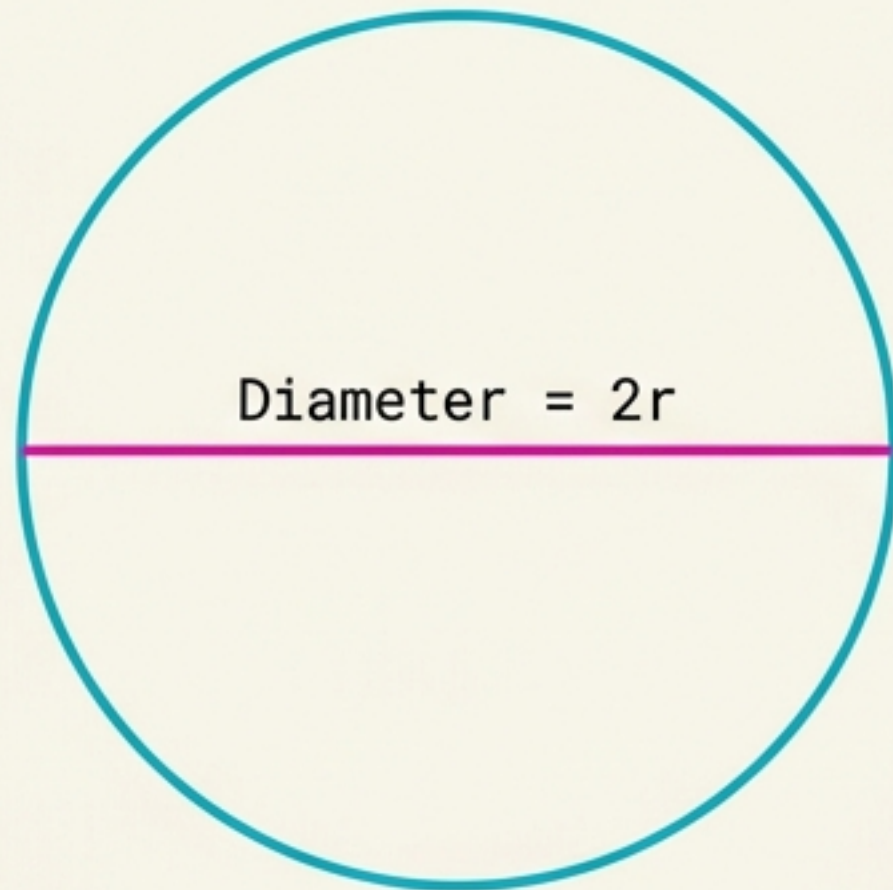
A Visual Guide to the 9-Dimensional Counterexample.



The Geometry of Diameter and Slicing

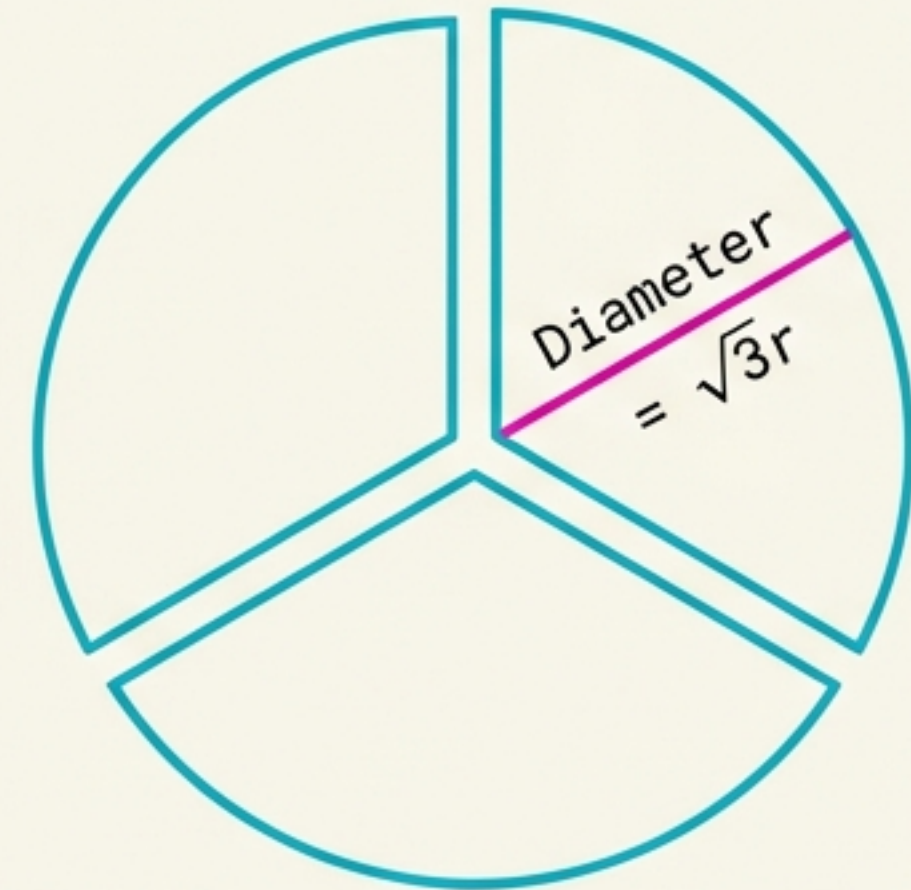
The Original Shape

Diameter: The largest possible distance between any two points in a set.



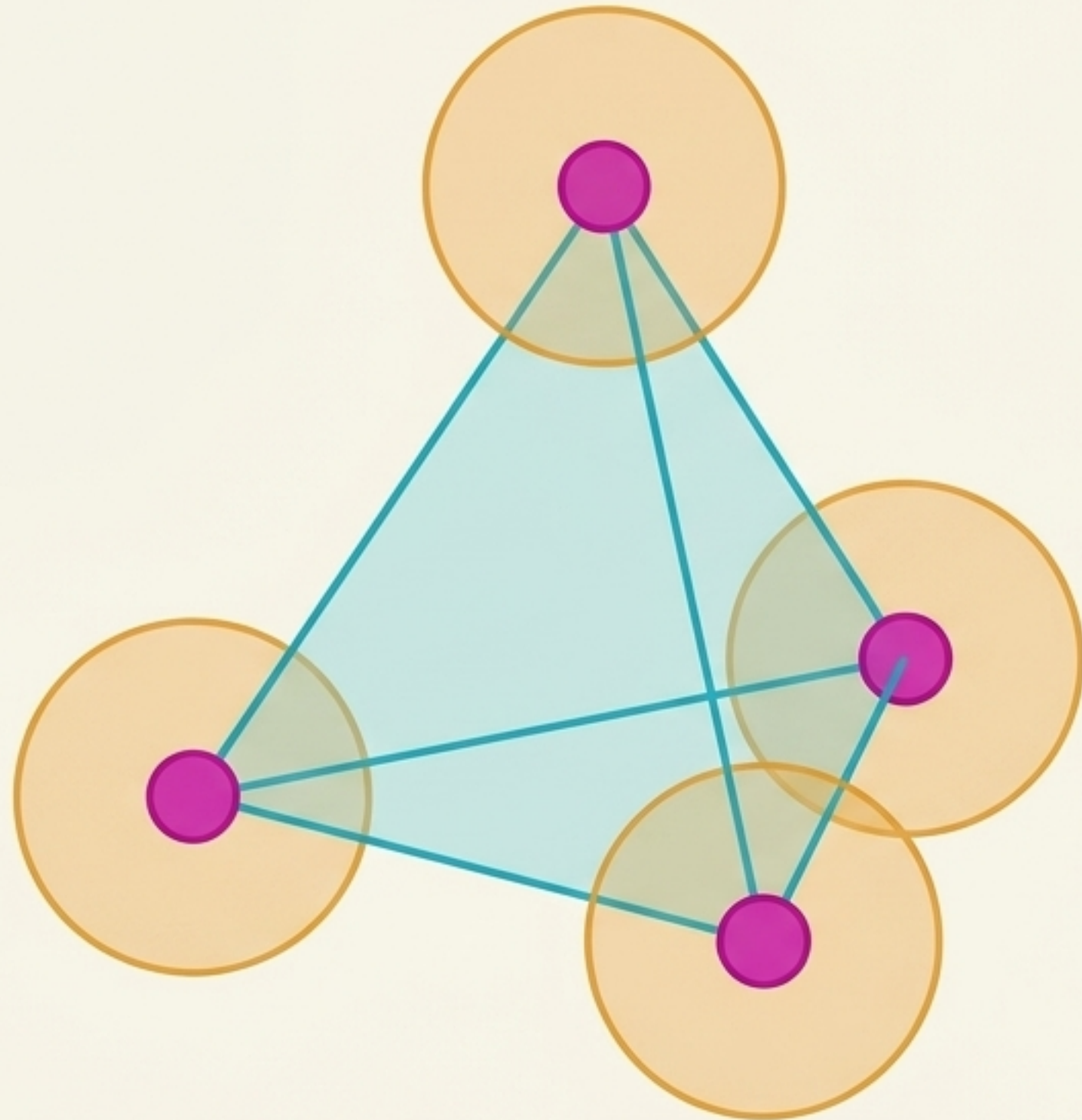
The Sliced Shape

Covering a shape with "smaller pieces" means every individual piece must have a strictly smaller diameter than the whole.



The shape is successfully covered by pieces of a strictly smaller diameter.

Borsuk's Question (1933)



In 1933, Polish mathematician Karol Borsuk asked a deceptively simple question:

“Can every bounded set in n -dimensional space ($\mathbb{R}^n$) be cut into $n+1$ pieces of strictly smaller diameter?”

Why exactly $n+1$? Look at a regular simplex (a triangle in 2D, a tetrahedron in 3D).

It has exactly $n+1$ corners. To make the pieces smaller than the original shape, no piece can contain two corners.

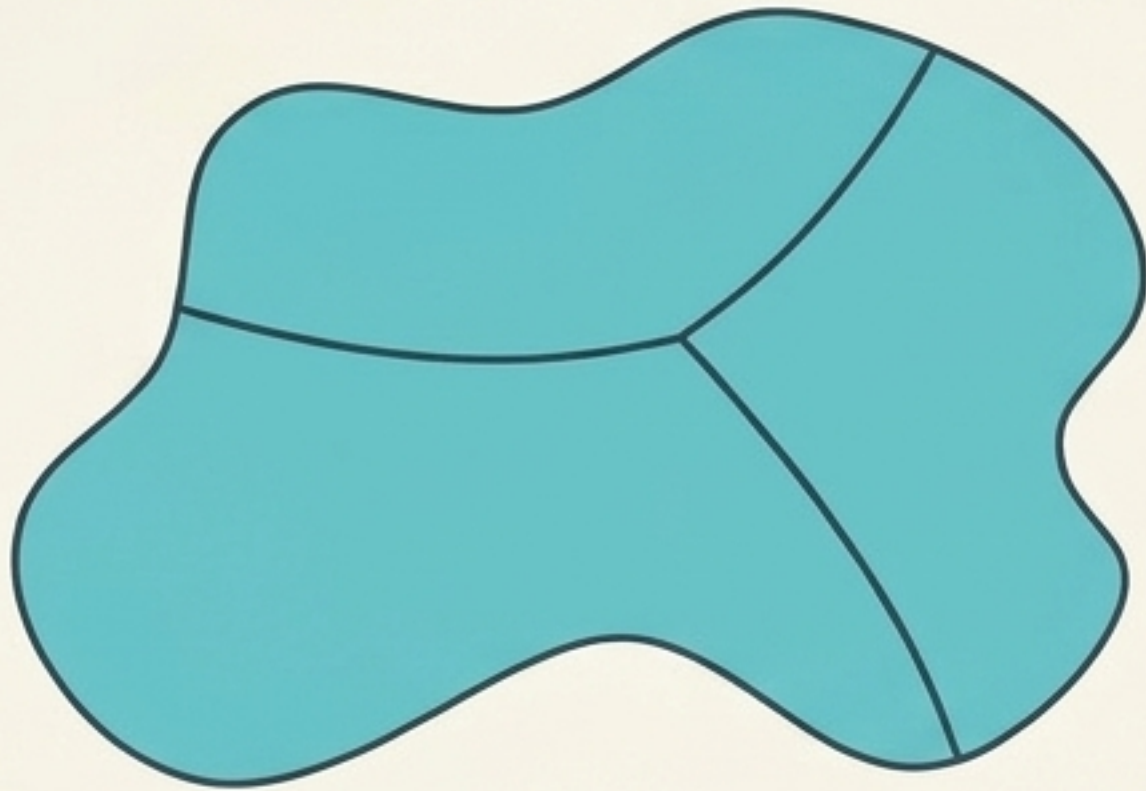
Therefore, $n+1$ pieces are the absolute minimum required.

Borsuk wondered if this was also the maximum ever needed.

Our 3D Intuition Says “Yes”

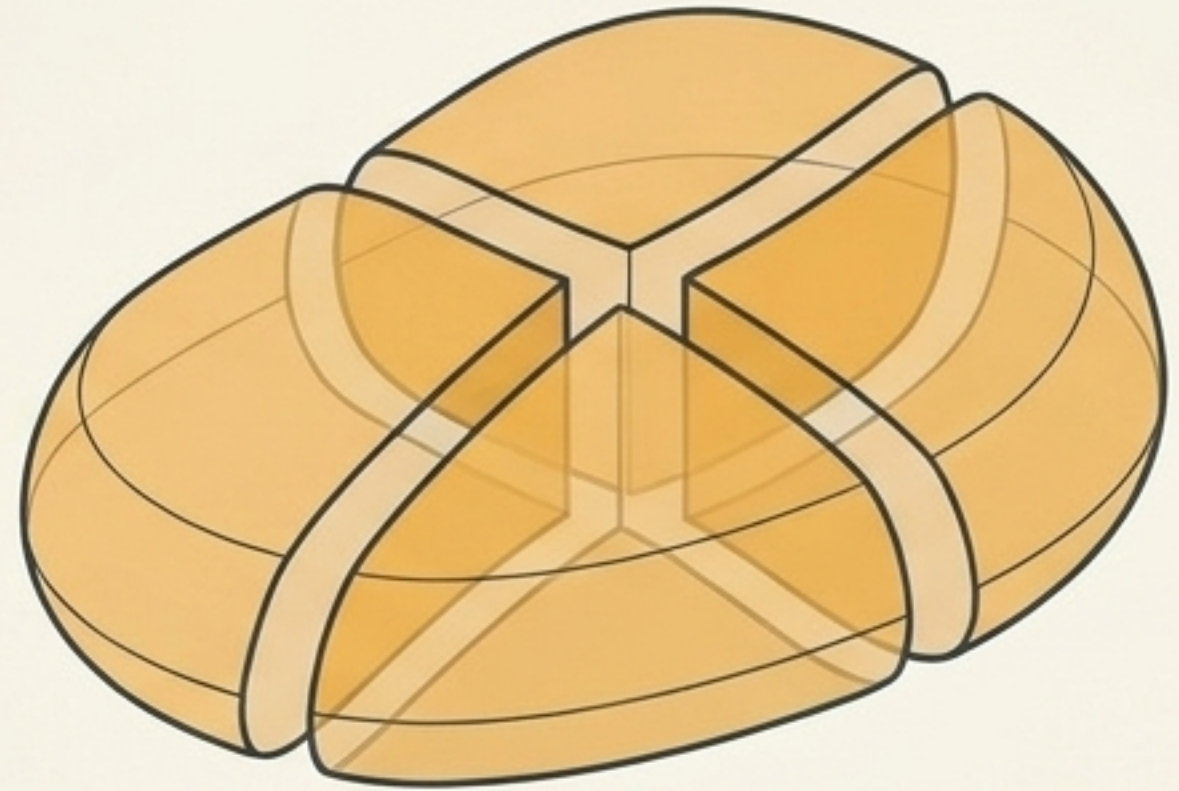
For decades, the mathematics community leaned toward believing the conjecture was a universal geometric truth. In the dimensions humans can easily visualize, Borsuk's rule holds perfectly.

Borsuk, 1932 (2D space)



Proved true for any 2D shape.

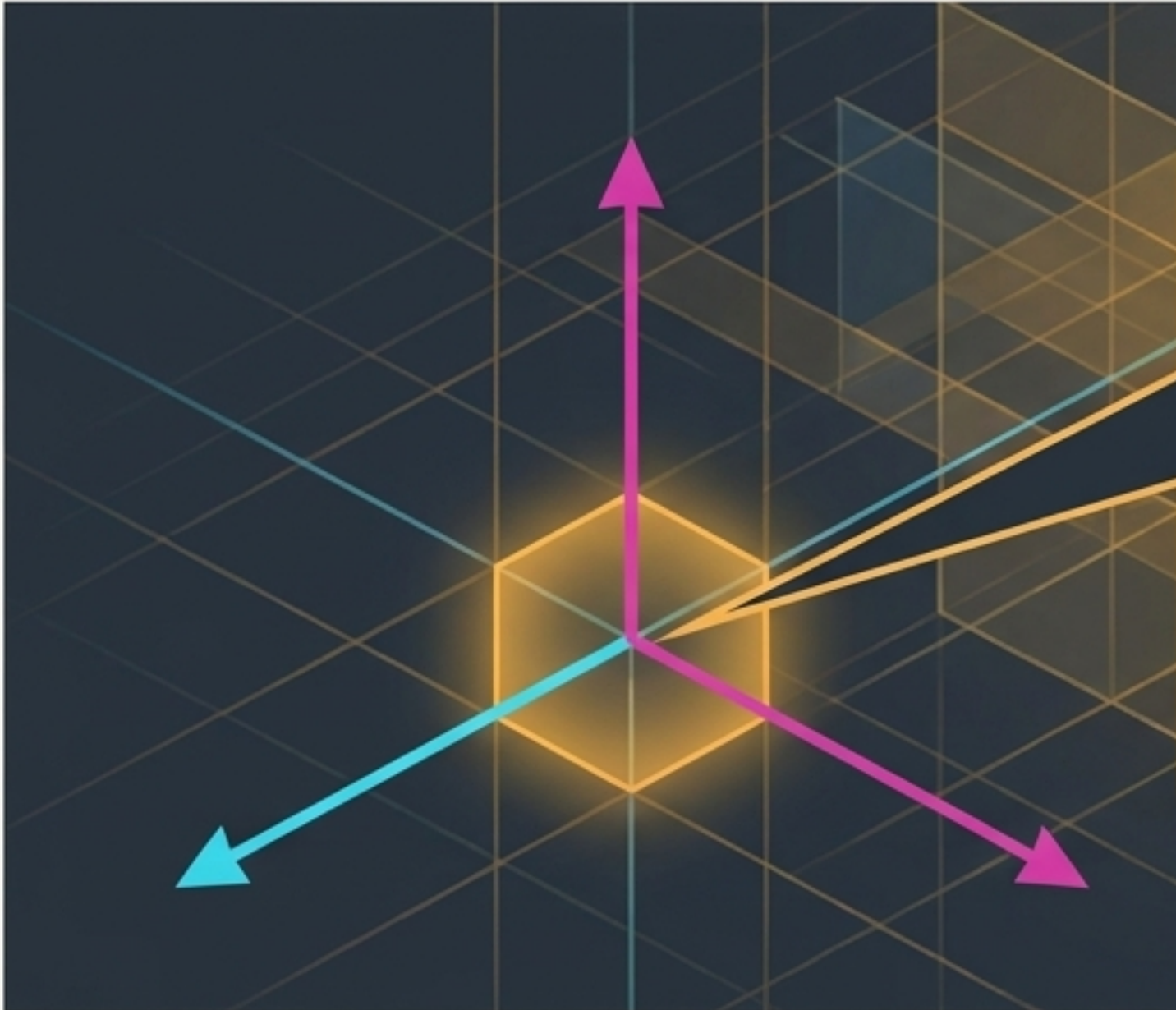
Perkal 1947, Eggleston 1955 (3D space)



Proved true for any 3D shape.

The 1993 High-Dimensional Disproof

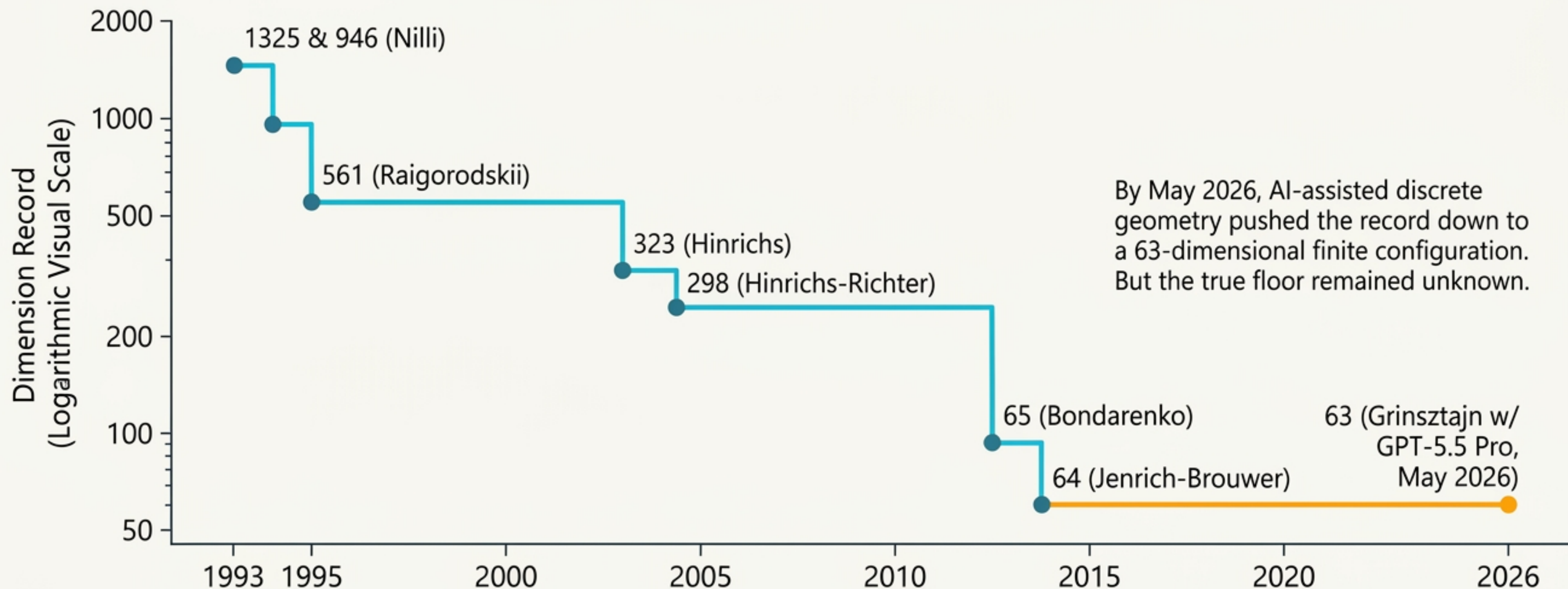
In 1993, Jeff Kahn and Gil Kalai shocked the math world by proving Borsuk wrong in high dimensions. Utilizing the Frankl-Wilson theorem, they constructed a counterexample using matrices formed by vectors with +1 and -1 coordinates.


$$|xx^T - yy^T|^2 = 2m^2 - 2\langle x, y \rangle^2$$

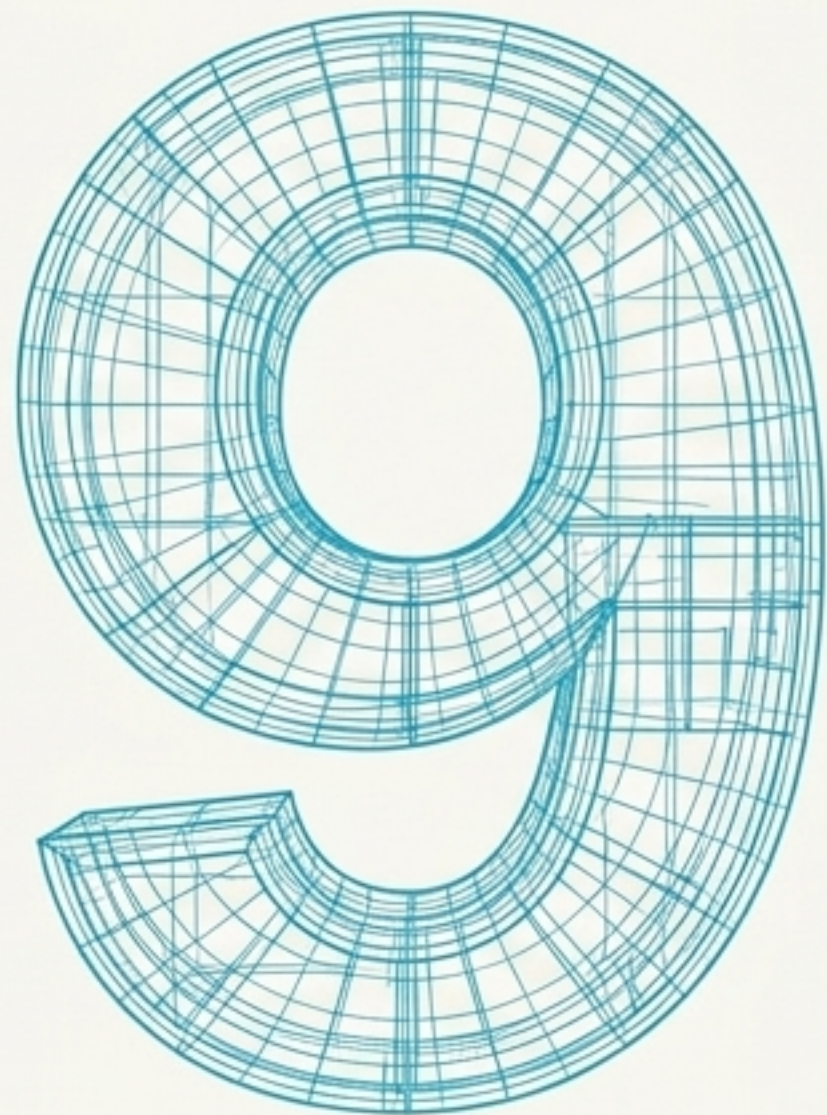
The Critical Insight: Because the dot product $\langle x, y \rangle$ is squared and subtracted, the distance between two points is maximized exactly when the original **vectors are perpendicular** ($\langle x, y \rangle = 0$). This geometric quirk forced high-dimensional shapes to require vastly more than $n+1$ pieces.

The 30-Year Race to the Bottom

Kahn and Kalai proved the conjecture failed for dimensions above 2014. This triggered a decades-long race to find the absolute lowest dimension where Borsuk's rule breaks.



The Threshold is Dimension 9



On September 23, 2026, an OpenAI preprint finally located the floor. Instead of a finite set of points, they built a continuous, infinite geometric object that shatters the conjecture in exactly 9 dimensions.

Blueprint

Theorem 1.1

The compact set X cannot be covered by ten sets of smaller diameter, meaning Borsuk's conjecture fails in dimension 9.

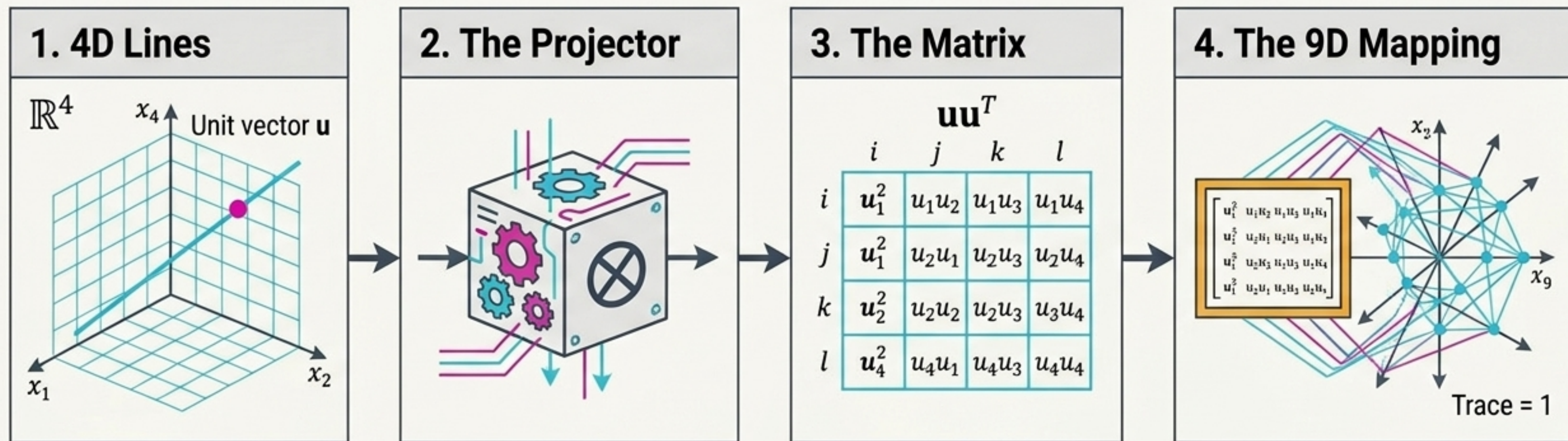
Blueprint

Corollary 7.1

Through a geometric extension, the conjecture therefore fails in EVERY dimension $d \geq 9$.

Constructing the 9D Counterexample (Set X)

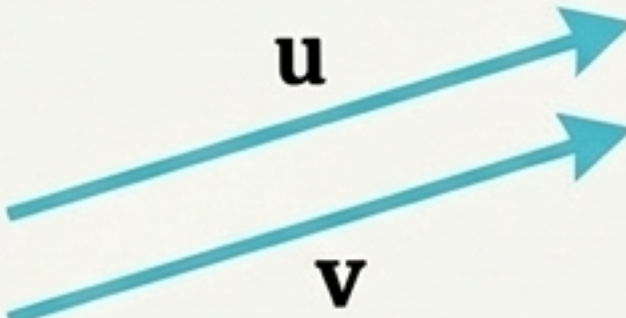
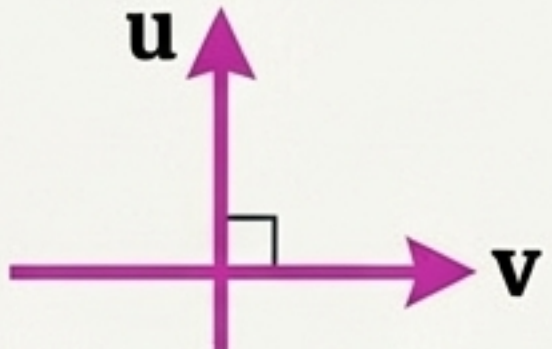
How do you build a 9-dimensional shape?



By stripping out redundant information, this continuous collection of symmetric 4x4 matrices perfectly maps into a 9-dimensional Euclidean space. This is our shape, $\mathbf{X}$.

The Magic Distance Formula

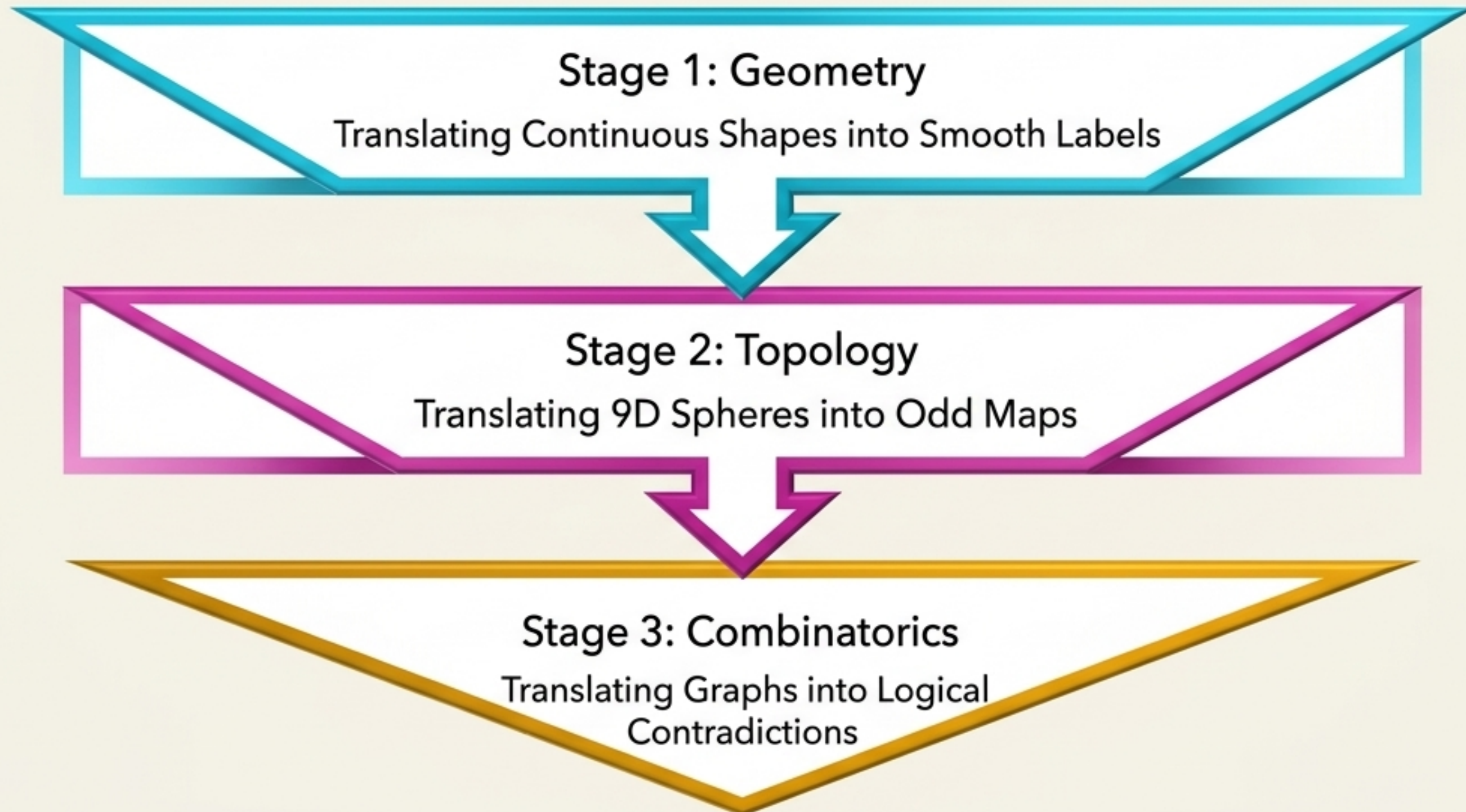
To prove Borsuk wrong, we must measure the distance between any two points in $\mathbf{X}$. Using the Frobenius distance, the formula is: $\sqrt{(2 - 2\langle \mathbf{u}, \mathbf{v} \rangle^2)}$

The Parallel Case Two parallel vectors		$\langle \mathbf{u}, \mathbf{v} \rangle^2 = 1$	Distance = 0
The Perpendicular Case Two perpendicular vectors		$\langle \mathbf{u}, \mathbf{v} \rangle^2 = 0$	Maximum Distance = $\sqrt{2}$

Because the dot product is squared, the maximum possible distance between any two points in $\mathbf{X}$ occurs exactly when their original 4D lines are perpendicular. This sets the absolute diameter of $\mathbf{X}$ at exactly $\sqrt{2}$.

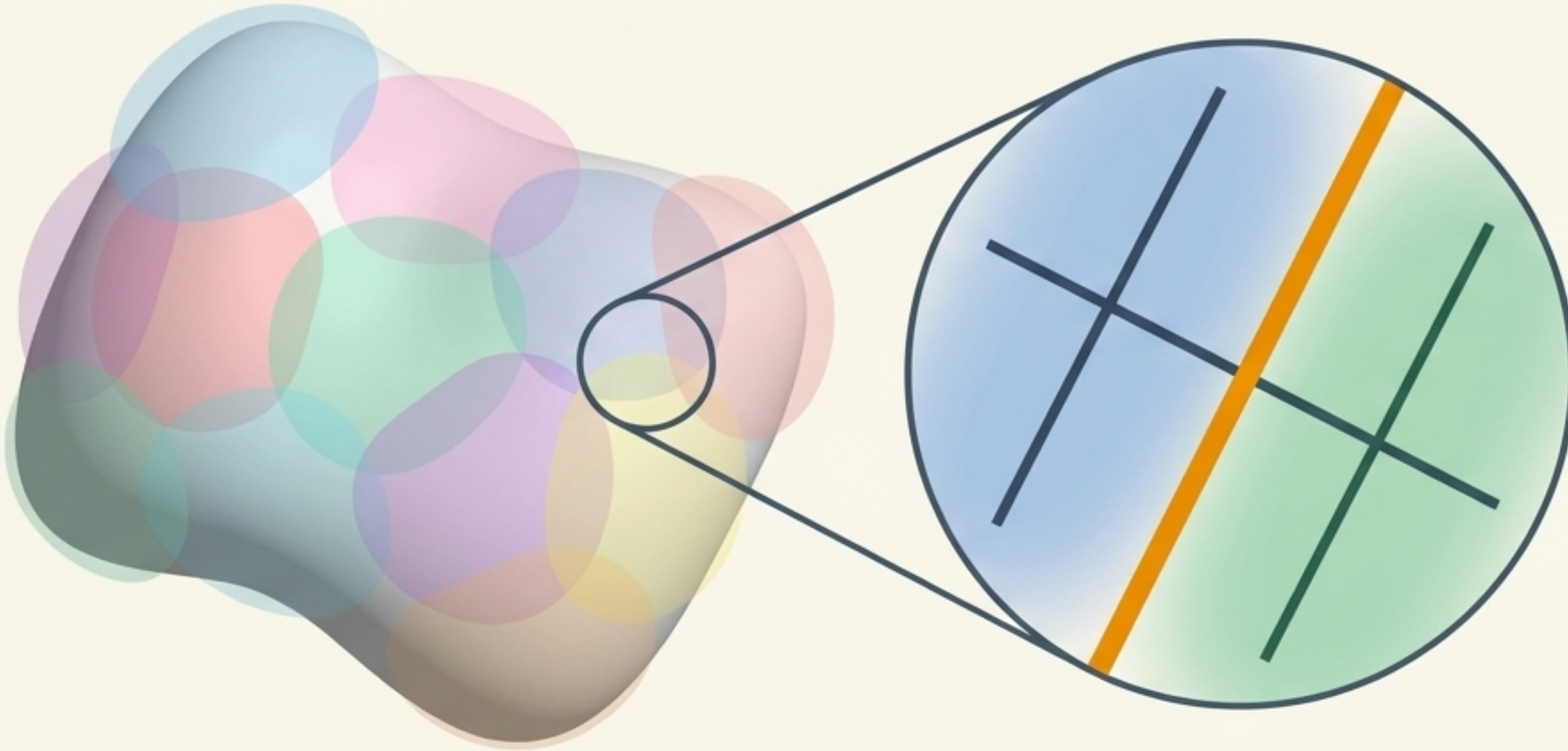
Proving the Impossible: A Three-Stage Architecture

Borsuk predicts that our 9D shape X can be covered by 10 smaller pieces. To prove this is impossible, the OpenAI paper executes a brilliant translation maneuver. It converts the geometric problem into a topological problem, and finally into a discrete logic puzzle that mathematically self-destructs.



Stage 1: Geometry to Smooth Labels

We begin by assuming Borsuk is right: imagine covering X with 10 sets of diameter strictly less than $\sqrt{2}$. We can smooth these sets out into 10 continuous functions, assigning “labels” to the surface.

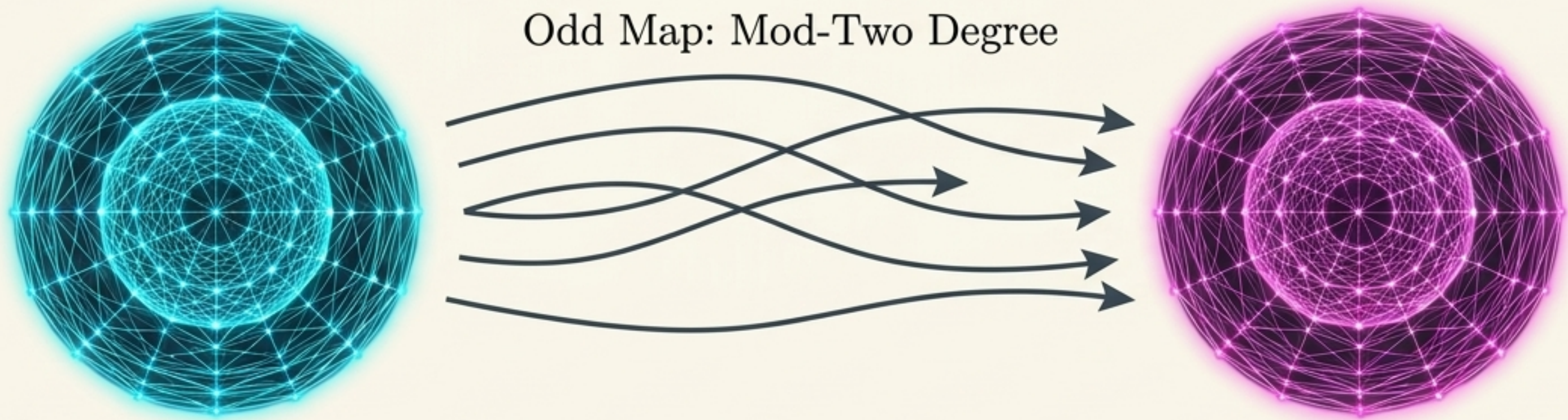


The Crucial Rule:

Because the pieces must be smaller than the maximum distance $\sqrt{2}$, and perpendicular lines are exactly $\sqrt{2}$ apart, perpendicular lines can never share a label.

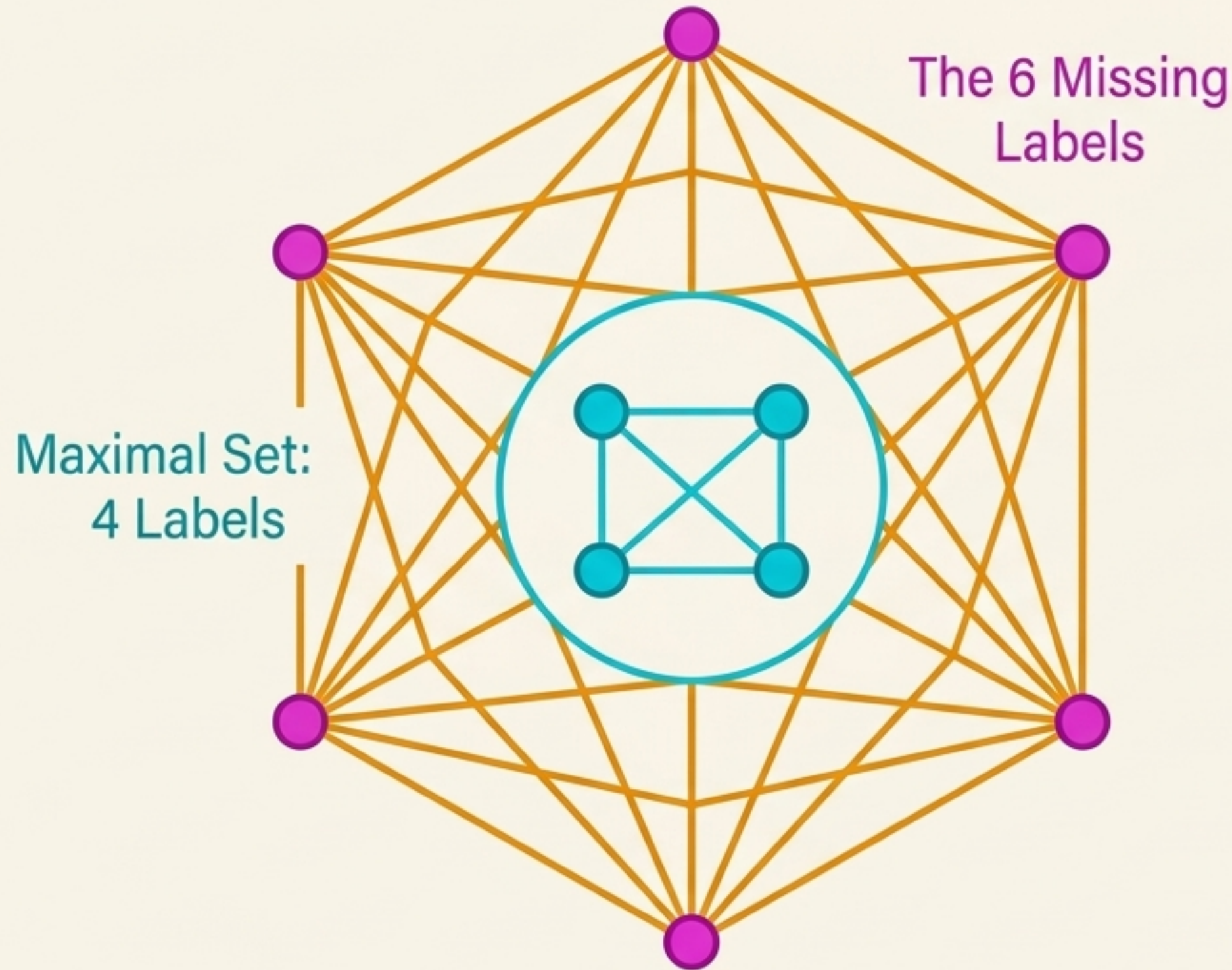
Stage 2: The Topological Obstruction

The paper extends these labels to all symmetric matrices, mapping the positive and negative parts of a matrix to opposite outputs. This creates an 'odd map' between 9-spheres.



The rules of topology dictate that a continuous odd map like this **MUST** hit **every single point** in the target space. This proves mathematically that exactly 10 labels is the absolute minimum, and every maximal label set must contain exactly 4 labels.

Stage 3: Combinatorics & Triangle Systems



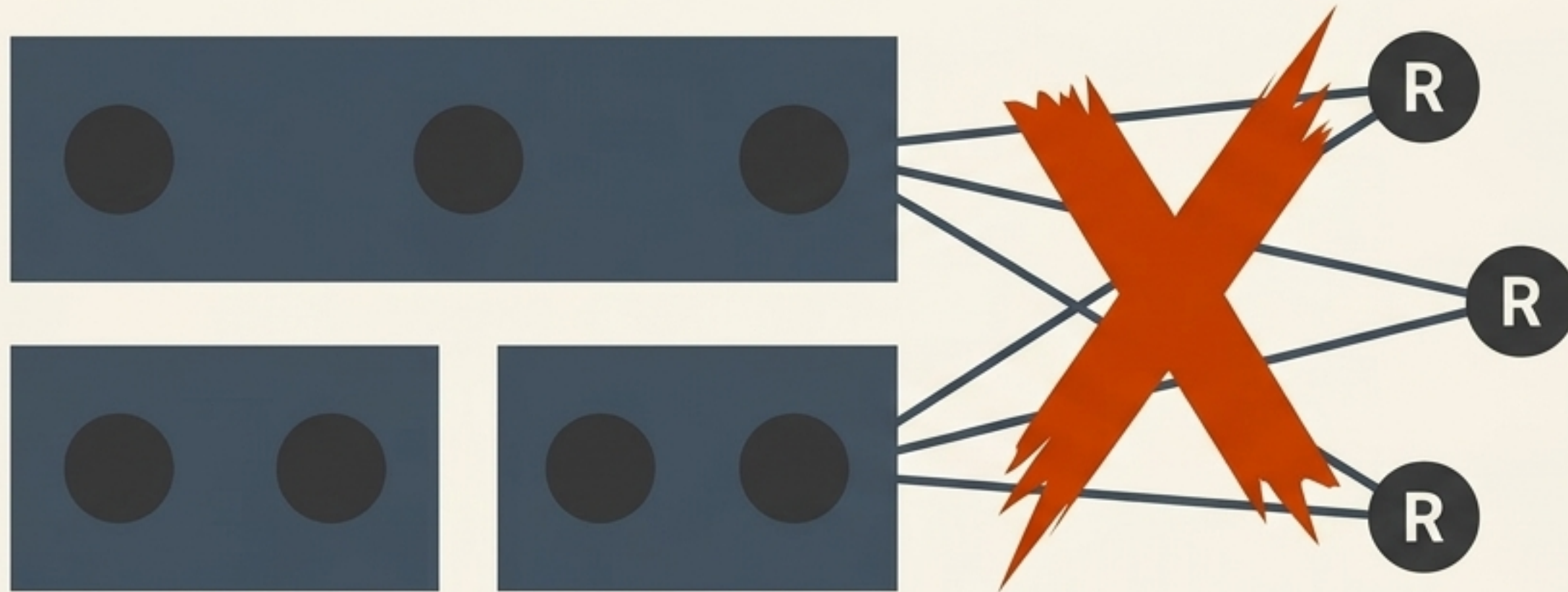
The geometry has been entirely stripped away, leaving a purely finite puzzle.

Because every maximal set must have exactly 4 labels, there are 6 missing labels from any given set.

The topological rules from Stage 2 force these 6 missing labels to form a highly rigid, interconnected graph called a "six-label triangle system." These systems behave in perfectly predictable ways as you traverse between adjacent sets of 4 labels.

The Final Contradiction: The 3-2-2 Impossibility

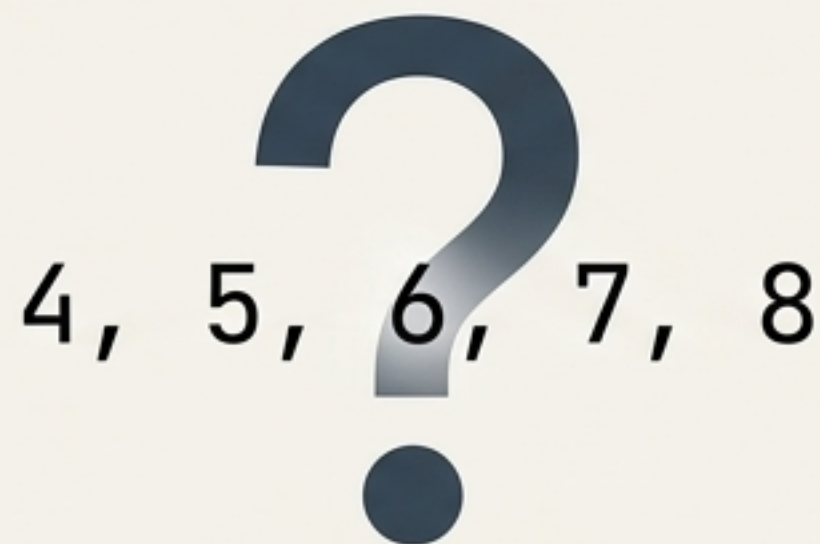
When researchers followed the strict combinatorial rules of these triangle systems across the entire network of labels, it forced the labels to organize into strict "partner blocks". The math dictates these blocks must settle into site sizes of 3, 2, and 2.



The **Checkmate**: A purely finite argument proves that a 3, 2, 2 block structure contains a logical contradiction—it cannot physically exist. Therefore, the **10-piece cover cannot** exist. Borsuk's conjecture falls in 9D.

The Edge of Knowledge

The Uncharted Zone



Borsuk holds true in 1–3D, and fails for ≥ 9 D. Dimensions 4 through 8 remain total mysteries.

Exact Pieces



$$b(X) > 10$$

We know X needs more than 10 pieces to be covered, but the exact number it requires is still completely unknown.

Ironclad Math

```
✓ BorsukNine.lean  
theorem borsuk_nine  
  (X : Set ℝ^9)  
  : b(X) > 9  
  :=  
  by sorry
```

While the paper is currently an unreviewed preprint, the main theorem has been completely formalized and machine-verified in the Lean programming language. The proof is undeniable.